

SKILLWEEK11

TASK1: To connect Modules, Validate Interfaces, Test End-to-End Execution, Handle Errors, Improve Reliability, Refactor Components

```
GNU nano 7.2 task1.c *
#ifndef CALCULATOR_H
#define CALCULATOR_H

int add(int a, int b);
int subtract(int a, int b);

#endif
```

```
GNU nano 7.2 task1.c *
#include "calculator.h"

int add(int a, int b)
{
    return a + b;
}

int subtract(int a, int b)
{
    return a - b;
}
```

```
GNU nano 7.2 main.c *
#include <stdio.h>
#include "calculator.h"

int main()
{
    int a = 10;
    int b = 5;

    printf("Addition = %d\n", add(a, b));
    printf("Subtraction = %d\n", subtract(a, b));

    return 0;
}
```

```
jhansi_priya@Jhansi-Priya: ~/ $ mkdir Skill11
jhansi_priya@Jhansi-Priya: ~/ $ cd Skill11
jhansi_priya@Jhansi-Priya: ~/Skill11$ nano calculator.h
jhansi_priya@Jhansi-Priya: ~/Skill11$ nano calculator.c
jhansi_priya@Jhansi-Priya: ~/Skill11$ nano main.c
jhansi_priya@Jhansi-Priya: ~/Skill11$ gcc main.c calculator.c -o
program
jhansi_priya@Jhansi-Priya: ~/Skill11$ ./program
Addition = 15
Subtraction = 5
jhansi_priya@Jhansi-Priya: ~/Skill11$ |
```

TASK2: To detect Runtime Errors, Generate Messages, Log Failures, Handle Invalid Syntax, Recover Gracefully, Test Error Scenarios

```
GNU nano 7.2 error_handling.c *
#include <stdio.h>

int main()
{
    int a, b;

    printf("Enter two integers: ");

    if (scanf("%d %d", &a, &b) != 2)
    {
        printf("ERROR: Invalid input. Please enter integers.\n");
        return 1;
    }

    if (b == 0)
    {
        printf("ERROR: Division by zero is not allowed.\n");
        return 1;
    }

    printf("Result = %d\n", a / b);

    return 0;
}
```

```
jhansi_priya@Jhansi-Priya:~/Skill11$ nano error_handling.c
jhansi_priya@Jhansi-Priya:~/Skill11$ gcc error_handling.c -o error_handling
jhansi_priya@Jhansi-Priya:~/Skill11$ ./error_handling
Enter two integers: 10 2
Result = 5
jhansi_priya@Jhansi-Priya:~/Skill11$ ./error_handling
Enter two integers: 10 4
Result = 2
jhansi_priya@Jhansi-Priya:~/Skill11$ ./error_handling
Enter two integers: 10 0
ERROR: Division by zero is not allowed.
jhansi_priya@Jhansi-Priya:~/Skill11$ ./error_handling
Enter two integers: abc xyz
ERROR: Invalid input. Please enter integers.
jhansi_priya@Jhansi-Priya:~/Skill11$ ./error_handling > output.log
cat output.log
jhansi_priya@Jhansi-Priya:~/Skill11$ cat output.log
Enter two integers: ERROR: Invalid input. Please enter integers.
jhansi_priya@Jhansi-Priya:~/Skill11$ ./error_handling > output.log
cat output.log
Enter two integers: Result = 5
jhansi_priya@Jhansi-Priya:~/Skill11$ ./error_handling 2> error.log
cat error.log
Enter two integers: 8 4
Result = 2
```

TASK3: To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings

```
GNU nano 7.2 memory.c *
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *arr;

    arr = malloc(5 * sizeof(int));
    if (arr == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < 5; i++)
    {
        arr[i] = i + 1;
    }

    printf("Values:\n");
    for (int i = 0; i < 5; i++)
    {
        printf("%d ", arr[i]);
    }

    printf("\n");

    return 0;
}
```

```
jhansi_priya@Jhansi-Priya: ~/Skill11$ nano memory.c
jhansi_priya@Jhansi-Priya:~/Skill11$ gcc -g memory.c -o memory
jhansi_priya@Jhansi-Priya:~/Skill11$ ./memory
Values:
1 2 3 4 5
jhansi_priya@Jhansi-Priya:~/Skill11$ valgrind --leak-check=full ./memory
==662== Memcheck, a memory error detector
==662== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==662== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==662== Command: ./memory
==662==
Values:
1 2 3 4 5
==662==
==662== HEAP SUMMARY:
==662==   in use at exit: 20 bytes in 1 blocks
==662==   total heap usage: 2 allocs, 1 frees, 1,044 bytes allocated
==662==
==662== 20 bytes in 1 blocks are definitely lost in loss record 1 of 1
==662==   at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_m
emcheck-amd64-linux.so)
==662==   by 0x1091BE: main (memory.c:8)
==662==
==662== LEAK SUMMARY:
==662==   definitely lost: 20 bytes in 1 blocks
==662==   indirectly lost: 0 bytes in 0 blocks
==662==   possibly lost: 0 bytes in 0 blocks
==662==   still reachable: 0 bytes in 0 blocks
==662==   suppressed: 0 bytes in 0 blocks
==662==
```

```
    {
        arr[i] = i + 1;
    }

    printf("Values:\n");
    for (int i = 0; i < 5; i++)
    {
        printf("%d ", arr[i]);
    }

    printf("\n");
    free(arr);
    return 0;
}
```

```

j
jhansi_priya@Jhansi-Priya:~/Skill11$ nano memory.c
jhansi_priya@Jhansi-Priya:~/Skill11$ gcc -g memory.c -o memory
jhansi_priya@Jhansi-Priya:~/Skill11$ valgrind --leak-check=full ./memory
==673== Memcheck, a memory error detector
==673== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==673== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==673== Command: ./memory
==673==
Values:
1 2 3 4 5
==673==
==673== HEAP SUMMARY:
==673==   in use at exit: 0 bytes in 0 blocks
==673== total heap usage: 2 allocs, 2 frees, 1,044 bytes allocated
==673== All heap blocks were freed -- no leaks are possible
==673== For lists of detected and suppressed errors, rerun with: -s
==673== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
jhansi_priya@Jhansi-Priya:~/Skill11$ valgrind --leak-check=full ./memory
> valgrind_report.txt 2>&1
jhansi_priya@Jhansi-Priya:~/Skill11$ cat valgrind_report.txt
==674== Memcheck, a memory error detector
==674== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==674== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==674== Command: ./memory

```

TASK4: To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate Reports

```

GNU nano 7.2 test.sh *
#!/bin/bash

echo "==== TEST 1 ===="
echo "10 2" | ./error_handling

echo ""
echo "==== TEST 2 ===="
echo "10 0" | ./error_handling

echo ""
echo "==== TEST 3 ===="
echo "abc xyz" | ./error_handling

echo ""
echo "==== TESTING COMPLETED ===="

```

```

jhansi_priya@Jhansi-Priya:~/Skill11$ nano test.sh
jhansi_priya@Jhansi-Priya:~/Skill11$ chmod +x test.sh
jhansi_priya@Jhansi-Priya:~/Skill11$ ./test.sh
==== TEST 1 ====
Enter two integers: Result = 5

==== TEST 2 ====
Enter two integers: ERROR: Division by zero is not allowed.

==== TEST 3 ====
Enter two integers: ERROR: Invalid input. Please enter integers.

==== TESTING COMPLETED ====

```


